

Problem

The voltage across a 2-H inductor is $20(1 - e^{-2t})$ V. If the initial current through the inductor is 0.3 A, find the current and the energy stored in the inductor at $t = 1$ s.

Step-by-step solution

Step 1 of 4

Consider the following information:

The voltage across 2 H inductor is, $v(t) = 20(1 - e^{-2t})$ V

The initial current through the inductor is, $i(0) = 0.3$ A

Write the expression for current through the inductor.

$$i = \frac{1}{L} \int_{t_o}^t v(t) dt + i(t_o)$$

At $t_o = 0$ s, the expression becomes as follows:

$$i = \frac{1}{L} \int_0^t v(t) dt + i(0)$$

Step 2 of 4

Substitute 2 H for L , $20(1 - e^{-2t})$ V for $v(t)$ and 0.3 A for $i(0)$.

$$\begin{aligned} i(t) &= \frac{1}{2} \int_0^t 20(1 - e^{-2t}) dt + 0.3 \\ &= \frac{20}{2} \left(\int_0^t 1 dt - \int_0^t e^{-2t} dt \right) + 0.3 \\ &= 10 \left(t - \frac{e^{-2t}}{-2} \right) \bigg|_0^t + 0.3 \\ &= 10 \left(t - \frac{e^{-2t} - 1}{-2} \right) + 0.3 \\ &= 10t + 5e^{-2t} - 5 + 0.3 \\ &= 10t + 5e^{-2t} - 4.7 \text{ A} \end{aligned}$$

Therefore, the expression for the inductor current is $i(t) = 10t + 5e^{-2t} - 4.7$ A.

Step 3 of 4

Find the inductor current at $t = 1$ s.

$$\begin{aligned} i(1 \text{ s}) &= 10(1) + 5e^{-2(1)} - 4.7 \\ &= 10 + 5(0.1353) - 4.7 \\ &= 5.977 \text{ A} \end{aligned}$$

Therefore, the current through inductor at $t = 1$ s is 5.977 A.

Step 4 of 4

Write the expression for energy stored in the inductor.

$$w_L = \frac{1}{2} L i^2$$

Here,

L is the value of inductor

i is the current through inductor.

Substitute 2 H for L and 5.977 A for i to find energy at $t = 1$ s.

$$\begin{aligned} w_L &= \frac{1}{2} (2) (5.977)^2 \\ &= 35.72 \text{ J} \end{aligned}$$

Therefore, the energy stored in the inductor at $t = 1$ s is 35.72 J.